

Total No. of Questions : 4]

SEAT No. :

PC196

[6361]-54

[Total No. of Pages :4

B.E. (Mechanical) (Insem)

HEATING VENTILATION AIR CONDITIONING &
REFRIGERATION

(2019 Pattern) (Semester- VII) (402041)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2; Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of Steam Tables, Mollier charts and electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Explain required physical and chemical properties of refrigerant. [6]

- b) An aircraft refrigeration plant has to handle a cabin load of 30 tonnes. The atmospheric temperature is 17°C. The atmospheric air is compressed to a pressure of 0.95 bar & temperature of 30°C due to ram action. This air is then further compressed in a compressor to 4.75 bar, cooled in a heat exchanger to 67°C, expanded in turbine to 1 bar pressure & supplied to the cabin. The air leaves the cabin at a temperature of 27°C. The isentropic efficiencies of both compressor & turbine are 0.9. [9]

Determine the following,

- i) Mass flow rate of air circulated/Seconds
- ii) COP
- iii) Specific power required.

Take $C_p = 1.004$ kJ/kg-K and $\gamma = 1.4$ for air

Sketch the cycle on T-S diagram.

OR

P.T.O.

Q2) a) Explain with neat sketch the regenerative air refrigeration system. [6]

- b) A bootstrap cooling system of 10 TR capacity is used in an airplane. The ambient air pressure and temperature are 20°C and 0.85 bar respectively. The pressure of air increases from 0.85 bar to 1 bar due to ramming action of air. The pressure of air discharged from the main compressor is 3 bar. The discharge pressure of air from the secondary compressor is 4 bar. The isentropic efficiency of each compressor is 80%, while that of turbine is 85%. The temperature drop of air in the first and second heat exchanger is 87.85°C and 88.3°C respectively. Assuming the ramming action to be isentropic, the required cabin pressure is of 0.9 bar and temperature of 20°C . Take C_p of air as 1 and $\gamma = 1.4$. [9]

Find

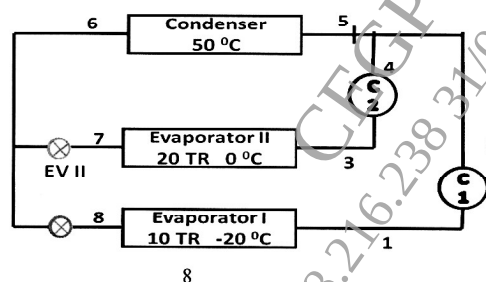
- Draw neat temperature entropy plot of given system.
- The power required to operate the system.
- The C.O.P. of the system.

Q3) a) Explain the demerits of using a single compressor for a large pressure ratio in refrigeration systems. [5]

- b) A multi evaporator system with individual compressor and an individual expansion valve using R 22 as refrigerant as shown in figure below. Assuming exit of condenser to be saturated liquid and entry to each compressor to be saturated vapour, [10]

Find

- Total power required to run the system in kW.
- C.O.P. of system and
- Draw neat Log P Vs h plot on the R 22 chart provided and attach as supplement.



OR

- Q4)** a) Explain two fluids cascade cycle with the help of schematic and p-h diagram. [5]
- b) Explain following multi-pressure systems with the help of schematic and p-h diagram. Also write formula for COP of the systems. [10]
- i) Two-evaporator system with single compressor and individual expansion valves.
 - ii) Two-evaporator system with individual compressor and multiple expansion valve.



